

TRIASHA SARKAR

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PhD student in Aerospace Engineering at Georgia Tech (ASDL) working on machine learning and statistical modeling for sustainable transportation and engineering systems, with an emphasis on uncertainty quantification. Prior industry experience in airworthiness certification at Rolls-Royce.

EDUCATION

- PhD, Aerospace Engineering — Georgia Institute of Technology, Atlanta, GA** May 2025 – Present
Anticipated graduation: **[MM/YY — FILL IN BEFORE SUBMITTING]**
- Graduate Research Assistant, Aerospace Systems Design Laboratory (ASDL).
 - Graduate coursework: ISYE 6739 Statistical Methods (A); ISYE 6414 Regression Analysis (Summer 2026).
- MSc, Aerospace Vehicle Design — Cranfield University, United Kingdom** Mar 2022 – Feb 2023
First Class Honours · Research on hybrid propulsion, liquid hydrogen, alternative fuels, and environmental noise mitigation.
- B.Tech, Aerospace Engineering — SRM Institute of Science and Technology, India** Jul 2017 – Jun 2021
First Class Honours.

TECHNICAL SKILLS

Programming Languages: Python, C++, MATLAB
ML & Scientific Computing: PyTorch, scikit-learn, NumPy, SciPy, pyDOE2, NetworkX, GeoPandas, OSMnx, Jupyter, Git/GitHub
Methods: Neural networks, CNNs, Gaussian process regression, time-series anomaly detection, clustering, feature extraction, surrogate and reduced-order modeling, Operator Inference, proper orthogonal decomposition, uncertainty quantification, regression and statistical inference, design of experiments
Domain Modeling: Life-cycle assessment (LCA), techno-economic analysis (TEA), greenhouse-gas and carbon-intensity modeling, network resilience and accessibility modeling, parametric vehicle sizing
Engineering Tools: NASTRAN, PATRAN, ANSYS

SELECTED RESEARCH

- Machine Learning for Silent Failure Detection in Simulation Campaigns** May 2026 – Present
Georgia Tech (AE8900) · Advised by Dr. Adam Cox
- Found that **45% of a 15,120-case solid rocket motor design sweep produced no output and no error message** — silent failures invisible to standard validity checks.
 - Built a leakage-aware classification pipeline predicting failure directly from geometry at **96.8% accuracy (AUC 0.986)**, combining time-series outlier detection, clustering, CNN-based feature extraction, and phase-aware preprocessing.
 - Traced the root cause to an uncapped convergence loop in the solver source, converting a statistical finding into a reproducible software defect.
- Atlanta Mobility Resilience Digital Twin** 2026 – Present
Independent project · github.com/triasha72/atlanta-mobility-resilience-digital-twin
- Prototype digital twin of Atlanta's road network (OSMnx, NetworkX) modeling disruption scenarios and comparing travel-time and accessibility outcomes across origin-destination pairs.
 - Extending toward larger stress-test scenarios, access to essential services, and neighborhood-level equity and vulnerability layers.
- Sustainable Aviation Fuel Lifecycle & Techno-Economic Analysis (Project GREEN TEA)** Sep 2025 – Apr 2026
Georgia Tech ASDL / U.S. Endowment for Forestry
- Well-to-wake life-cycle and techno-economic assessment of sustainable aviation fuel from forest residue under parametric uncertainty; modeled energy pathways, feedstock logistics, greenhouse-gas emissions, and carbon intensity.
 - Benchmarked carbon-intensity estimates against **REET, CORSIA, and 40BSAF-REET** regulatory and life-cycle frameworks.
- NURBS-BEM Electromagnetic Solver with Learned Surrogates** Oct 2025 – Present
github.com/triasha72/NURBS_BEM_EMSolver
- Built a mesh-free boundary-element solver for solenoid magnetic field computation using NURBS geometry; generated parametric multifidelity datasets for surrogate modeling, POD feasibility analysis, and Operator Inference.
- Surrogate Model Learning: GP, RSM, and RBF Comparative Study** 2025 – Present
github.com/triasha72/Surrogate-model-learning
- Compared Gaussian process, response-surface, and radial-basis-function surrogates across benchmarks; characterized how approximation assumptions drive error, uncertainty behavior, and failure modes in nonlinear regimes.

Alternative-Configuration Aircraft under Climate Targets (FAA ASCENT 64)

May 2025 – Apr 2026

Georgia Tech ASDL

- Physics-based parametric modeling of alternative long-range aircraft configurations under uncertain CO₂ targets within the ICAO Long-Term Aspirational Goal (LTAG) framework.

Demand & Economic Modeling for a Lifting-Body Airship (Project EAGLE)

Sep 2025 – Apr 2026

Vehicle Grand Challenge: RIC Enterprise / Georgia Tech ASDL

- Parametric sizing and demand modeling for a 750+ passenger lifting-body airship, including fare-sensitivity and fleet-sizing analysis across uncertainty scenarios.

MSc Thesis: Supersonic Business Jet Noise Mitigation

Dec 2022 – Feb 2023

Cranfield University / SENECA consortium (Rolls-Royce, DLR, ONERA)

- Primary-nozzle geometry optimization for FAA FAR 91.817 noise compliance in a data-scarce, simulation-expensive design space.

INDUSTRY EXPERIENCE

Systems Engineer — Alten India Pvt Ltd, embedded at Rolls-Royce, Bangalore

Oct 2023 – Apr 2025

- Supported airworthiness certification of the Trent XWB-84 EP engine; validated engineering models against certification standards.
- Worked directly with the consequences of model failure in deployed systems — the experience that motivates current research on silent failure detection.

Summer Intern, Avionics — STAR, Surat, India

Jun 2021 – Jul 2021

- Avionics design with Arduino for solid-rocket-motor static-test safety and data acquisition.

Summer Intern, Structures — Bangalore Aircraft Industries Ltd

Jun 2019 – Jul 2019

- Structural stress and material analysis under environmental loading using NASTRAN and PATRAN.

LEADERSHIP & SERVICE

Graduate Chair, Women of Aeronautics and Astronautics (WoAA), Georgia Tech Chapter

2025 – Present